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Lecture 3: Behavioral games

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Behavioral/laboratory games

- What is the laboratory?
- Lab games as an approach to measurement
  - Context-free measures of "economic" preferences
  - "Empirical anthropology"
  - Types of things that may be measured
    - Risk preferences: risk aversion/neutrality/loving
    - Time preferences: patience, discounting
    - Social preferences: altruism, pro-social behavior, trust
    - Cognitive skills: behavior under cognitive load

Benefits of this approach to measurement

- Closer to real behavior (?) than survey experiments
- Fully controlled environment; removes other contextual information, social norms
- Useful for measuring heterogeneity
- Survey fatigue? Respondent engagement?

Terminology

- "rational-actor" framework;
  - "the action chosen by a decision-maker is at least as good, according to her preferences, as every other available action" (Osborne, 2004, p.6).
- "rational" preferences must be:
  - Complete; for any two actions, a person knows whether she prefers the outcome under action a to action b
  - Transitive; if a person prefers the outcome under action a to that under action b, and b to c, she must prefer a to c
  - Ordinal; only the relative, not the absolute, value of utility matters (doesn't matter how much more she prefers outcome under action a over b)
- What does it mean for players to be "self-interested?"
- Does rational action require self-interest?

Terminology

- One-shot vs. repeated play
- Subgame perfect equilibrium
- Strategy method vs. dynamic interactions

Some specific games

From Camerer and Fehr (2004)

Table 1: Seven experimental games useful for measuring social preferences

Game	Definition of the Game	Real life Example	Predictions with rational and selfish players	Experimental regularities, References	Interpretation
Prisoners' dilemma Game	Two players, each of whom can either cooperate or defect. Payoffs are as follows: Cooperate: H,H; S,T Defect: T,S; L,L H>L, T>H, L>S	Production of negative externalities (pollution, loud noise), exchange without binding contracts, status competition.	Defect	50% choose Cooperate. Communication increases frequency of cooperation Dawes (1980)**	Reciprocate expected cooperation
Public Goods Game	n players simultaneously decide about their contribution g_i. ($0 \le g_i \le y$) where y is players' endowment; each player *i* earns $\pi_i = y - g_i + mG$ where G is the sum of all contributions and $m<1<mn$.	Team compensation, cooperative production in simple societies, overuse of common resources (e.g., water, fishing grounds)	Each player contributes nothing, i.e. $g_i = 0$.	Players contribute 50% of y in the one-shot game. Contributions unravel over time. Majority chooses $g_i=0$ in final period. Communication strongly increases cooperation. Individual punishment opportunities greatly increase contributions. Ledyard (1995)**.	Reciprocate expected cooperation
Ultimatum Game	Division of a fixed sum of money S between a Proposer and a Responder. Proposer offers x. If Responder rejects x both earn zero, if x is accepted the Proposer earns S – x and the Responder earns x.	Monopoly pricing of a perishable good; "11th-hour" settlement offers before a time deadline	Offer $x=\varepsilon$; where ε is the smallest money unit. Any $x>0$ is accepted.	Most offers are between .3 and .5S. $x<.2S$ rejected half the time. Competition among Proposers has a strong x-increasing effect; competition among Responders strongly decreases x. Güth et al (1982)*, Camerer (in press)**	Responders punish unfair offers; negative reciprocity
Dictator Game	Like the ultimatum game but the Responder cannot reject, i.e., the "Proposer" dictates (S-x, x).	Charitable sharing of a windfall gain (lottery winners giving anony-mously to strangers)	No sharing, i.e., x = 0	On average "Proposers" allocate x=.2S. Strong variations across experiments and across individuals Kahneman et al (1986)*, Camerer (in press)**	Pure altruism

Trust Game	Investor has endowment S and makes a transfer y between 0 and S to the Trustee. Trustee receives 3y and can send back any x between 0 and 3y. Investor earns S – y + x. Trustee earns 3y – x.	Sequential exchange without binding contracts (buying from sellers on Ebay)	Trustee repays nothing: x = 0. Investor invests nothing: y = 0.	On average y = .5S and trustees repay slightly less than .5S. x is increasing in y. Berg et al (1995)*, Camerer (in press)**	Trustees show positive reciprocity.
Gift Exchange Game	"Employer" offers a wage *w* to the "worker" and announces a desired effort level $\hat{e}$. If worker rejects (w, $\hat{e}$) both earn nothing. If worker accepts, he can choose *any e* between 1 and 10. Then employer earns 10e – w and worker earn $w - c(e)$. $c(e)$ is the effort cost which is strictly increasing in *e*.	Noncontractibility or nonenforceability of the performance (effort, quality of goods) of workers or sellers.	Worker chooses e = 1. Employer pays the minimum wage.	Effort increases with the wage *w*. Employers pay wages that are far above the minimum. Workers accept offers with low wages but respond with e = 1. In contrast to the ultimatum game competition among workers (i.e., Responders) has no impact on wage offers. Fehr et al (1993)*	Workers reciprocate generous wage offers. Employers appeal to workers' reciprocity by offering generous wages.
Third Party Punishment Game	A and B play a dictator game. C observes how much of amount S is allocated to B. C can punish A but the punishment is also costly for C.	Social disapproval of unacceptable treatment of others (scolding neighbors).	A allocates nothing to B. C never punishes A.	Punishment of A is the higher the less A allocates to B. Fehr and Fischbacher (2001a)*	C sanctions violation of a sharing norm.

Note: ** denotes survey papers, * denotes papers that introduced the respective games.

Variations

- Ultimatum game w/ competition
- Opportunities for punishment
- Public goods game with partners vs. strangers

Ethical considerations

- Pay compared to time spent (1-2 days' wages...several months' wages?)
- Social consequences
- Other concerns?

Relevance

- Why is any of this relevant for political science/policy?
- How do you know you're measuring the "right" thing, and that respondents are interpreting the game the way you want them to?
- Can we compare across societies?

References I

Camerer, C. F. and Fehr, E. (2004). Measuring social norms and preferences using experimental games: A guide for social scientists. Foundations of human sociality: Economic experiments and ethnographic evidence from fifteen small-scale societies, 97:55–95.

Osborne, M. J. (2004). An Introduction to Game Theory, volume 3. Oxford University Press New York.